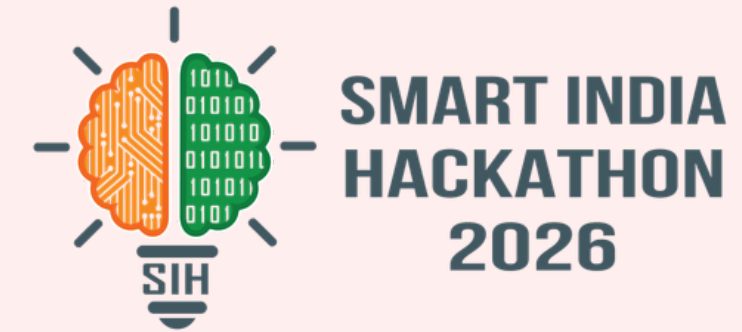




SMART INDIA HACKATHON 2026



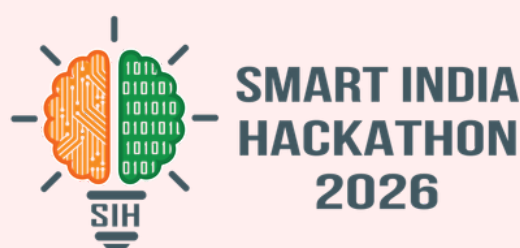
JOCKY

- **Problem Statement ID - 26148**
- **Problem Statement Title** - Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions
- **Theme - Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID-**
- **Team Name - SIG_OUT**





JOCKY - [TODO: Add one line text]

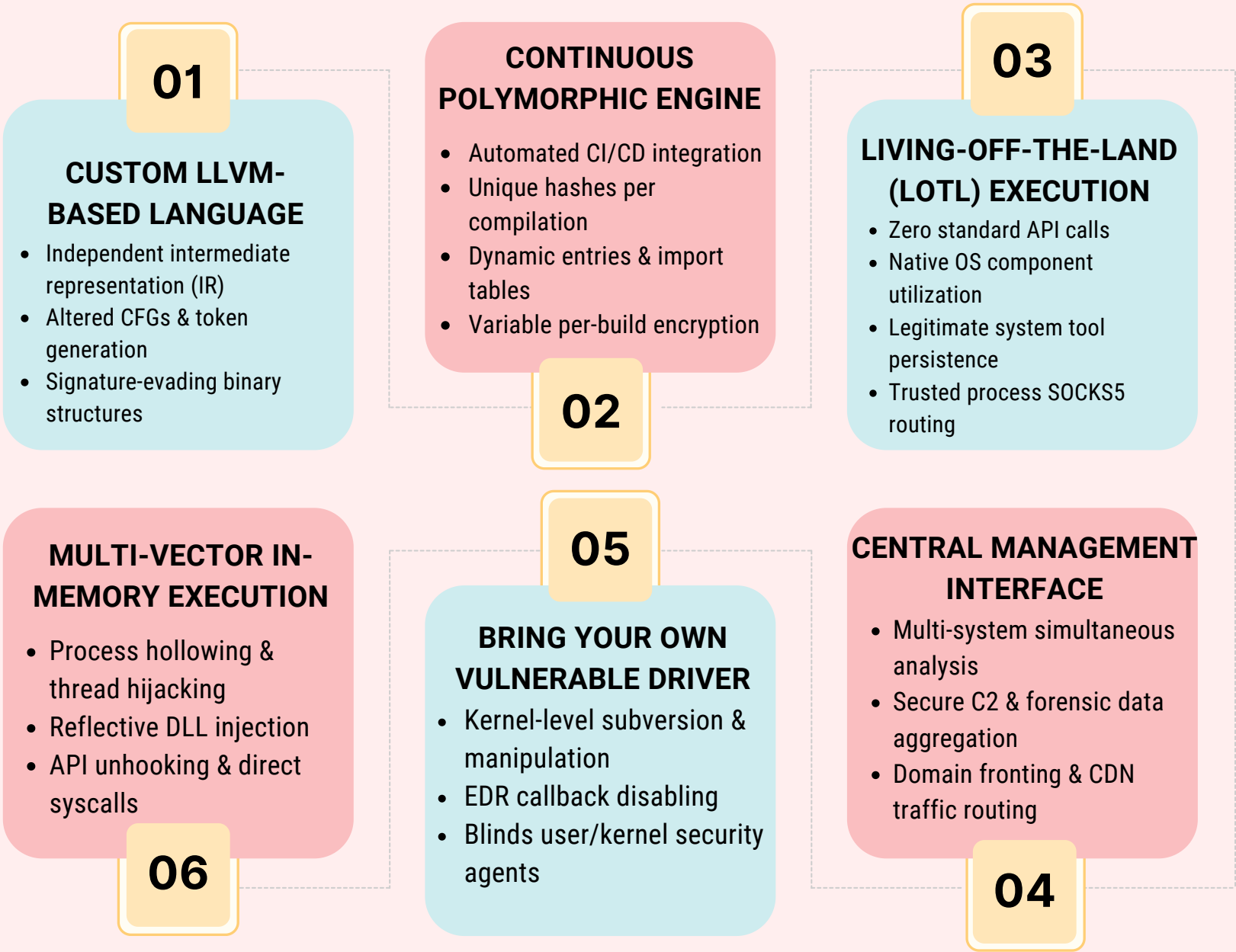


BRIEF OF IDEA -

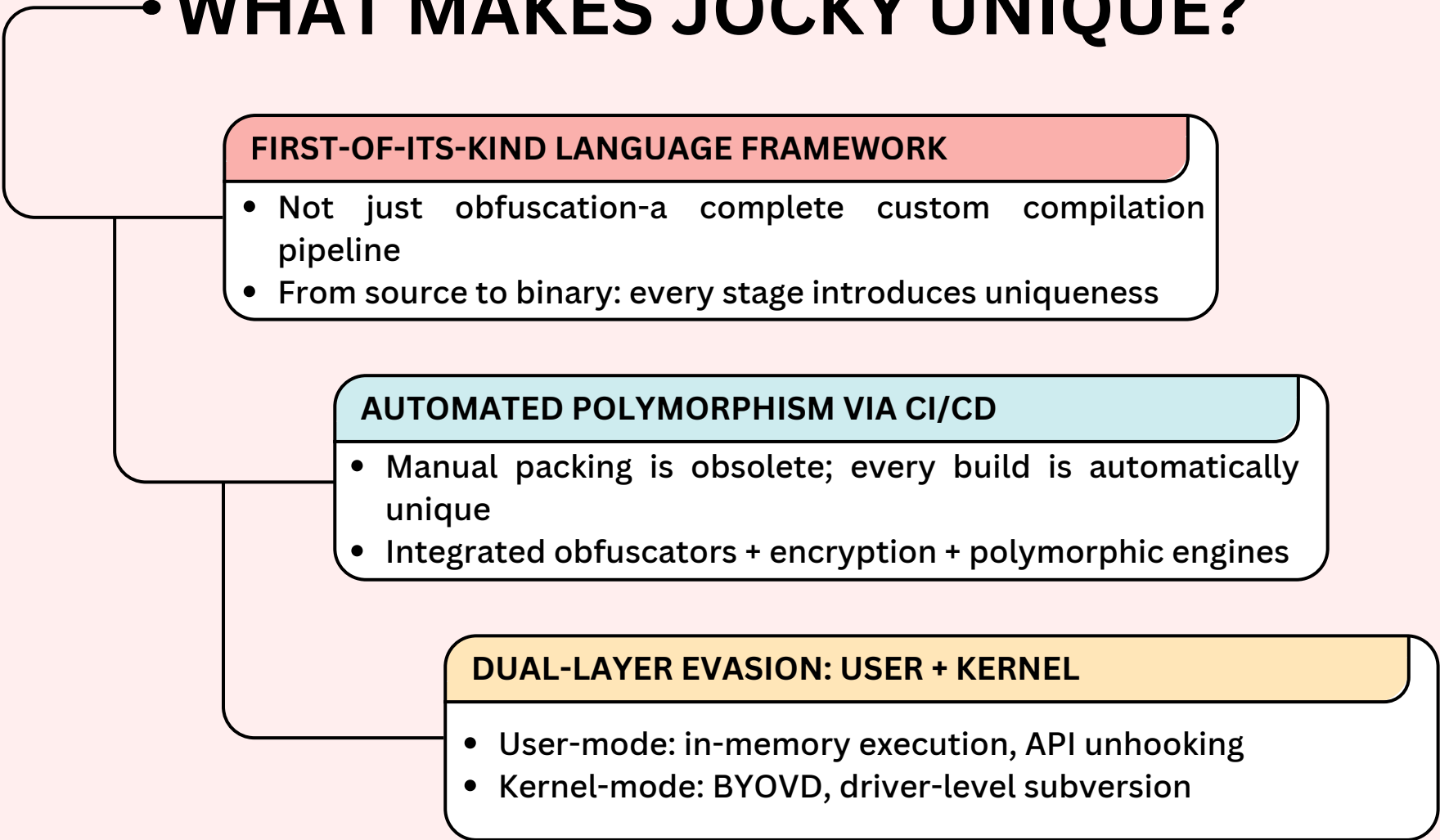
Creation of **forensic analysis scripts** in a **new language** to avoid antivirus detection and kernel level security by preventing file reputation and dynamic analysis through a **Continous Integration/Continous Deployment(CI/CD)** pipeline.

Proposed Solution

Jocky - A cross-platform programming language framework that enables systematic forensic analysis , bypassing all existing antivirus solutions through novel binary structures, polymorphic compilation, and living-off-the-land execution techniques.



WHAT MAKES JOCKY UNIQUE?



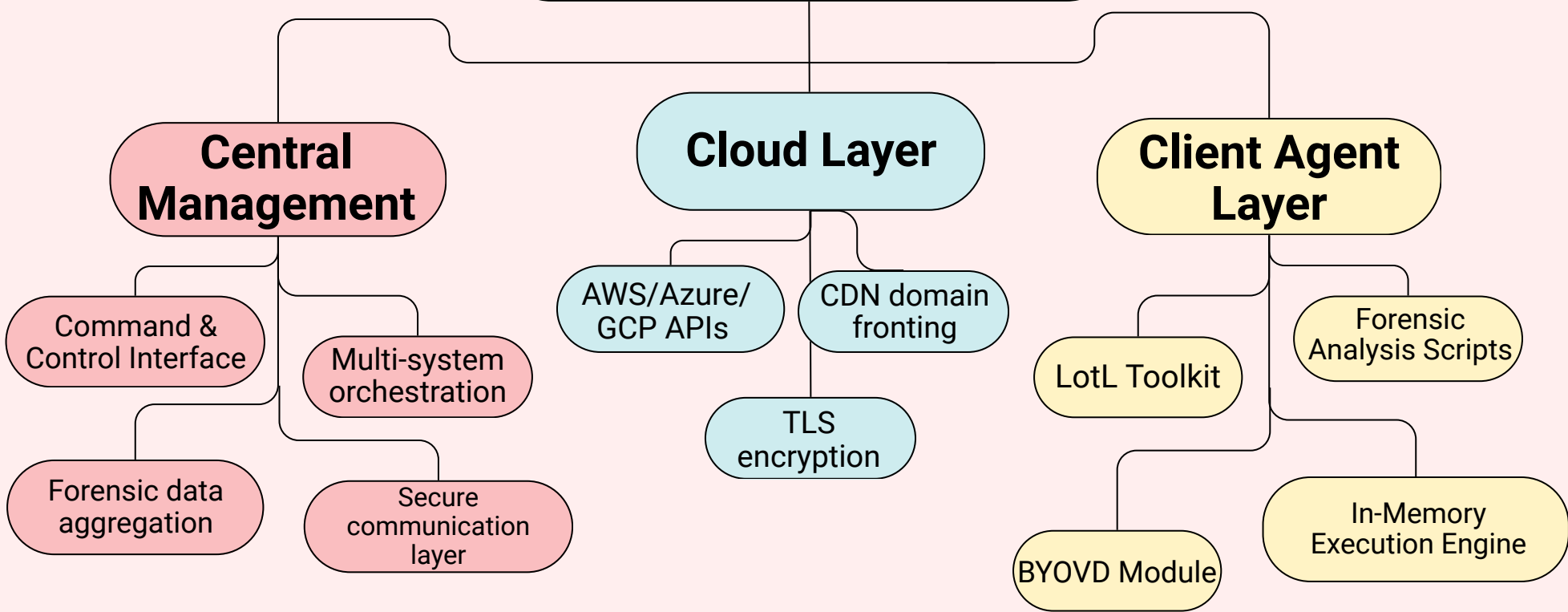
PROBLEMS ADDRESSED

- Custom LLVM IR alters binary signatures completely resulting in non detection by AV/EDR blocks
- LotL + direct syscalls bypass behavioral monitoring leading to aversion by behavioral heuristics methods
- Polymorphic engine generates unique binaries per deployment which tricks reputation databases maintained by antiviruses
- BYOVD techniques blind kernel security agents
- Domain fronting via trusted cloud infrastructure ensures that the network traffic does not get flagged

COMPILATION PIPELINE



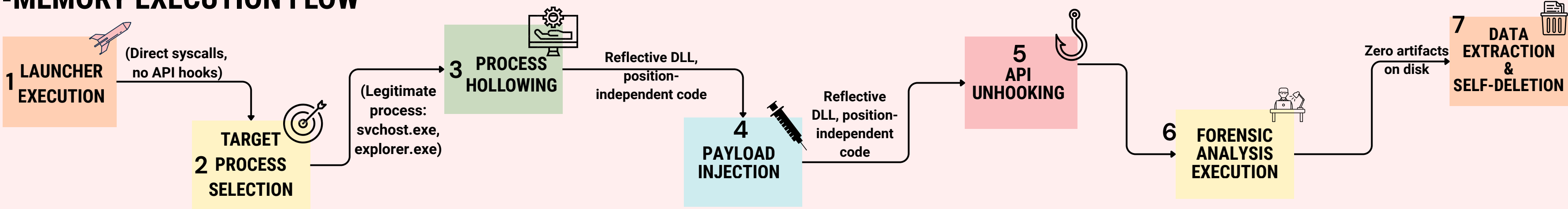
ARCHITECTURE



EVASION TECHNIQUES

LAYER	TECHNIQUE	MECHANISM	TARGET
STATIC	Custom LLVM IR Polymorphic Engine	Unique Binary Structures Hash/Entry Point Mutation	Signature DB File Reputation
BEHAVIORAL	Direct Syscalls API Unhooking	Bypass Hooked APIs Restore Original ntdll	EDR hooks Memory Scanners
MEMORY	Process Hollowing Reflective DLL	Inject into Trusted Process Load without LoadLibrary	Memory Scanners Import Table Analysis
KERNEL	BYOVD Driver Manipulation	Load Vulnerable Driver Disable EDR callbacks	Kernel Monitors Security Agents

IN-MEMORY EXECUTION FLOW



FEASIBILITY ANALYSIS

01 TECHNICAL FEASIBILITY

- LLVM compiler infrastructure is mature and extensible
- Cross-platform compilation (Windows/Ubuntu) is proven technology
- In-memory execution techniques are well-documented in research
- BYOVD methods leverage known vulnerable drivers (Microsoft WHQL signed)
- Domain fronting uses legitimate cloud infrastructure

02 ECONOMIC FEASIBILITY

- Open-source LLVM foundation (no licensing costs)
- Cloud infrastructure costs are pay-as-you-go (AWS/Azure)
- Development can leverage existing forensic tool research
- No specialized hardware required
- Scalable from single investigator to enterprise SOC teams

03 OPERATIONAL FEASIBILITY

- Cross-platform support covers 95% of enterprise environments
- Central management interface simplifies multi-system analysis
- Automated polymorphism reduces manual obfuscation effort
- Training required: forensic investigators with malware analysis background
- Integration with existing forensic workflows

POTENTIAL CHALLENGES & RISKS

CHALLENGE	IMPACT	LIKELIHOOD
AV vendors adapt to LLVM IR patterns	High	Medium
Microsoft patches vulnerable drivers (BYOVD)	High	High
Cloud providers block domain fronting	Medium	Low
Behavioral AI detects syscall patterns	Medium	Medium
Memory scanners detect hollowed processes	Medium	Medium

STRATEGIES FOR OVERCOMING

- Continuous LLVM IR Mutation:** Randomize IR generation patterns per compilation cycle
- Driver Rotation Pool :** Maintain database of 50+ vulnerable drivers, rotate automatically
- Multi-CDN Fallback:** Support AWS, Azure, GCP, Cloudflare domain fronting , CDN blocking
- Syscall Diversification:** Randomize syscall order, add junk syscalls, use indirect calls
- Multi-Process Injection:** Spread across 3-5 processes, not just one

DEVELOPMENT ROADMAP

LLVM FRONTEND	POLYMORPHIC ENGINE	EVASION MODULES	CENTRAL MANAGEMENT	TESTING AND HARDENING
• Custom IR generation • Control flow graph alteration • Token randomization	• Hash mutation pipeline • Entry point randomization • Import table modification	• In-memory execution engine • BYOVD framework • API unhooking	• C2 interface development • Multi-agent orchestration • Domain fronting integration	• Red team testing against 20+ AV/EDR • Performance omtimisation • Cross-platform validation
Phase 1	Phase 2	Phase 3	Phase 4	Phase 5

TARGET AUDIENCE IMPACT

Digital Forensic Investigators

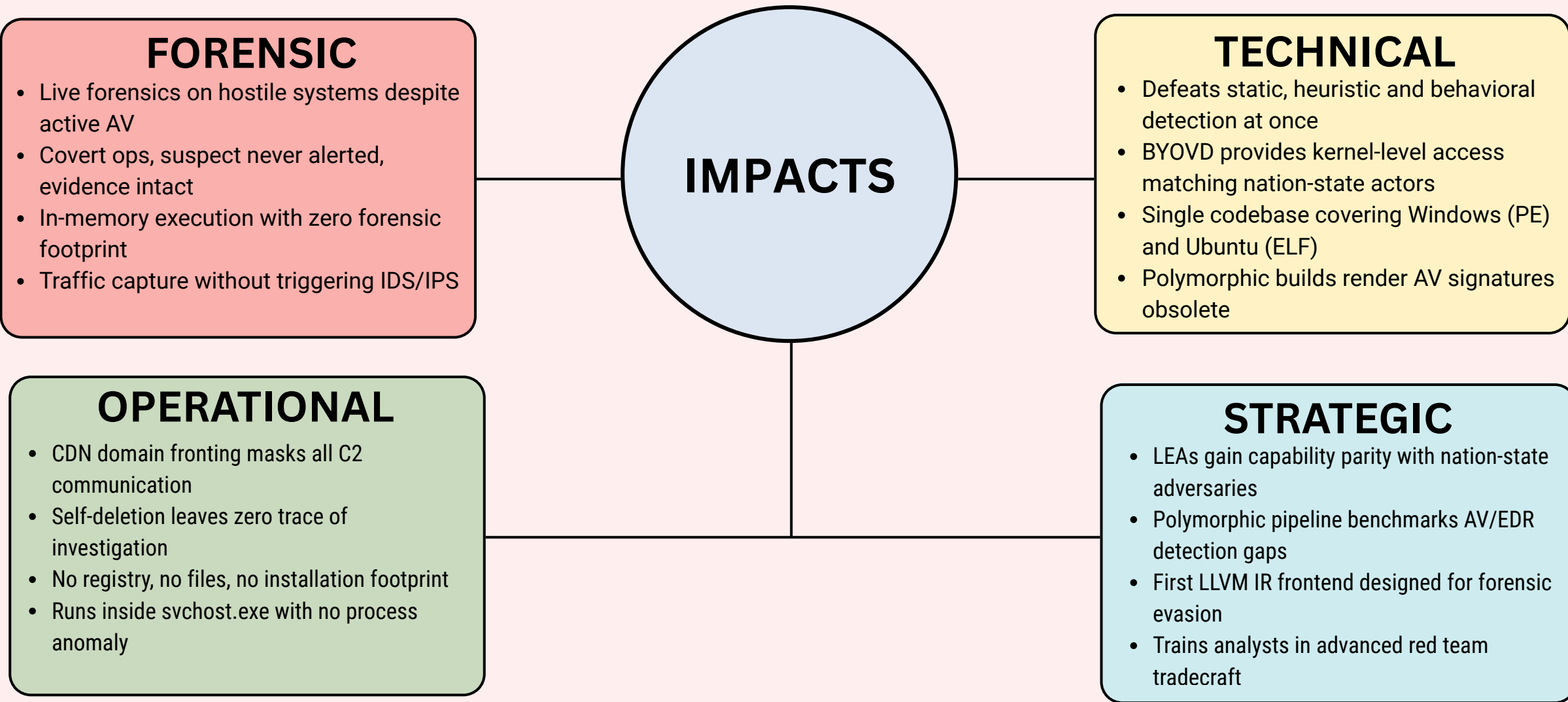
- Bypass AV restrictions on proprietary analysis tools
- Conduct deep system analysis without triggering alerts
- Analyze compromised systems in hostile environments
- Perform network forensics without detection
- Maintain operational security during investigations

Law Enforcement Agencies

- Investigate cybercrime without alerting suspects
- Conduct covert digital forensics in active investigations
- Analyze malware-infected systems safely
- Perform incident response on compromised networks
- Gather evidence without contamination

Enterprise Security Teams

- Simulate advanced adversaries for security testing
- Test EDR/AV effectiveness against novel techniques
- Validate detection capabilities against living-off-the-land attacks
- Assess incident response readiness
- Benchmark security posture against state-sponsored threats



APPLICATION SCENARIO PATHWAYS

Ransomware Investigation

- AV blocks analysis tools on live infected system
- JOCKY bypasses AV and analyzes ransomware behavior in-memory
- Decryption keys extracted with no suspect awareness

APT Investigation

- APT-deployed EDR blocks all standard forensic tooling
- BYOVD blinds EDR; JOCKY agents deployed covertly across network
- C2 infrastructure mapped without alerting threat actor

Insider Threat Investigation

- Standard forensic tools visibly alert the suspect
- Covert JOCKY deployment with zero footprint on suspect's system
- Evidence gathered silently with legally admissible chain maintained



RESEARCH AND REFERENCES



- Details / Links of the reference and research work

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.